Creating and Using the API Key


In Oracle Cloud Infrastructure (OCI), you need an **API key** to securely connect

- your external custom code (Python, etc.),
- local software development kits (SDKs), or
- the OCI Command Line Interface (CLI)

to your cloud account.

Why You Need It

- **Secure Authentication (No Passwords):** When writing code (such as a Python script calling the OCI Language service), passing your human login password over the web is highly insecure. Instead, OCI uses an asymmetric **RSA Key Pair** (a Private Key kept on your computer and a Public Key uploaded to OCI).

- **Cryptographic Signing:** Your local application code uses your private key to cryptographically "sign" every single API request it sends. When Oracle receives the request, it verifies the signature against the uploaded public key to confirm that the request genuinely came from you.

- **Identity & Permissions Enforcement:** The API key maps the code directly to your OCI User profile. This allows the Identity and Access Management (IAM) framework to confirm that you have the approved permissions to access resources, like extracting text or saving files to a bucket.

General Steps

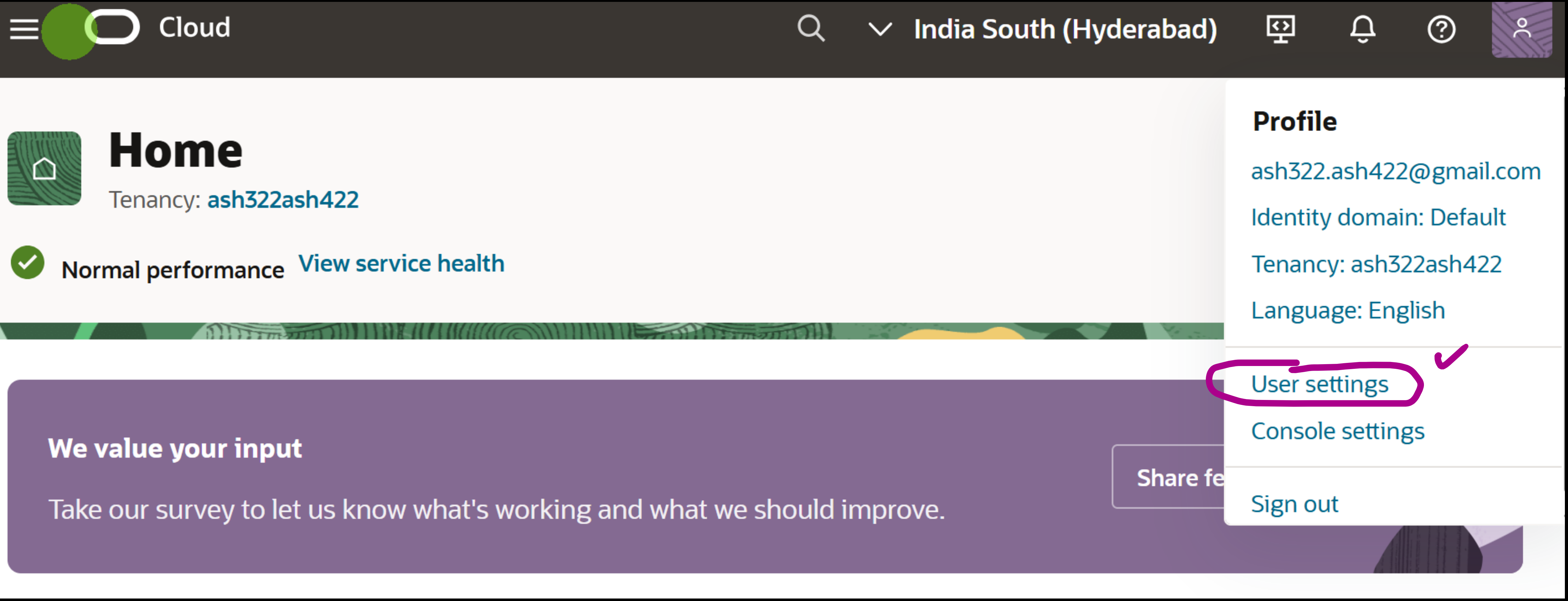
1) Navigate to Profile → User Settings → API Keys

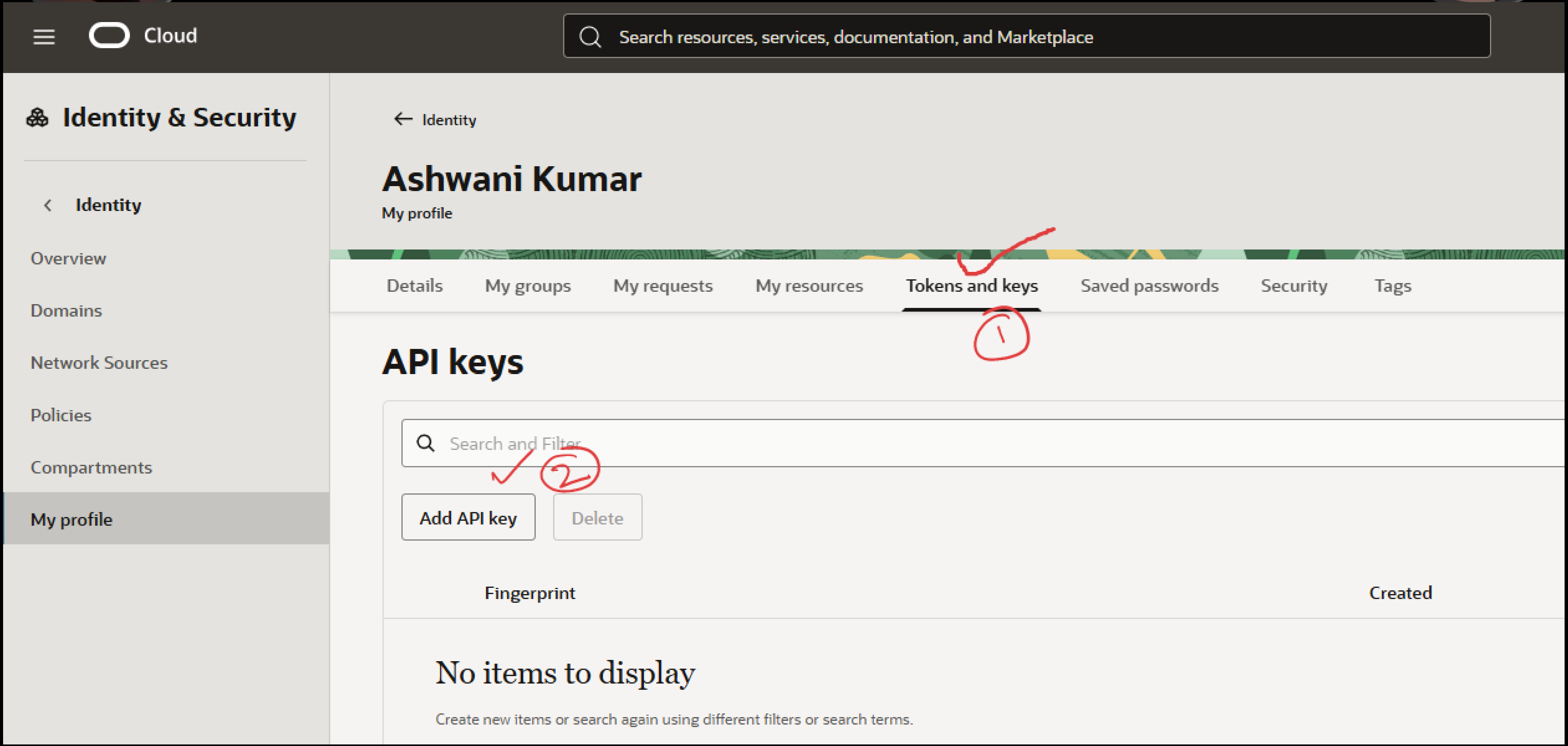
2) Generate -> API RSA Key Pair

3) Download the **private** key:
- Create a **config** file as described in the OCI instruction
- Your **private.pem** would get added here

4) Put above files in a special directory

5) Test the API key through the python code


Cloud
India South (Hyderabad)
Home
Tenancy: ash322ash422
Normal performance
View service health
Profile
ash322.ash422@gmail.com
Identity domain: Default
Tenancy: ash322ash422
Language: English
User settings
Console settings
Sign out
We value your input
Take our survey to let us know what's working and what we should improve.
Share fe


Cloud
Search resources, services, documentation, and Marketplace
Identity & Security
Identity
Overview
Domains
Network Sources
Policies
Compartments
My profile
Identity
Ashwani Kumar
My profile
Details
My groups
My requests
My resources
Tokens and keys
Saved passwords
Security
Tags
1
API keys
Search and Filter
2
Add API key
Delete
Fingerprint
Created
No items to display
Create new items or search again using different filters or search terms.

Add API key

Note: An API key is an RSA key pair in PEM format used for signing API requests. You can generate the key pair here and download the private key. If you already have a key pair, you can choose to upload or paste your public key file instead. Learn more

- (•) Generate API key pair ✓ 1
- ( ) Choose public key file
- ( ) Paste a public key

✓ 2 click here

Download private key

Download the private key. It will not be shown again. After you download it, change the file permissions so only you can view it

Download public key

3 This will download the API Key

4 click Add

Close | Add

Configuration file preview

Note: This configuration file snippet includes the basic authentication information you'll need to use the SDK, CLI, or other OCI developer tool. Paste the contents of the text box into your ~/.oci/config file and update the key_file parameter with the file path to your private key. Learn more

Fingerprint
c8:4c:f3:24:96:0a:94:e4:cf:78:49:36:39:7f:b1:2f

```
[DEFAULT]
user=ocid1.user.oc1..aaaaaaaaahu7lckuwznvwcebnv3gbodmbfl55hvcwyn7gxsmmaqzybq3kdwq
fingerprint=c8:4c:f3:24:96:0a:94:e4:cf:78:49:36:39:7f:b1:2f
tenancy=ocid1.tenancy.oc1..aaaaaaaafqfzlefu4wbgeutjopgcagugqrxyng4pn5m2ft6uoccpbd675uuq
region=ap-hyderabad-1
key_file=<path to your private keyfile> # TODO
```

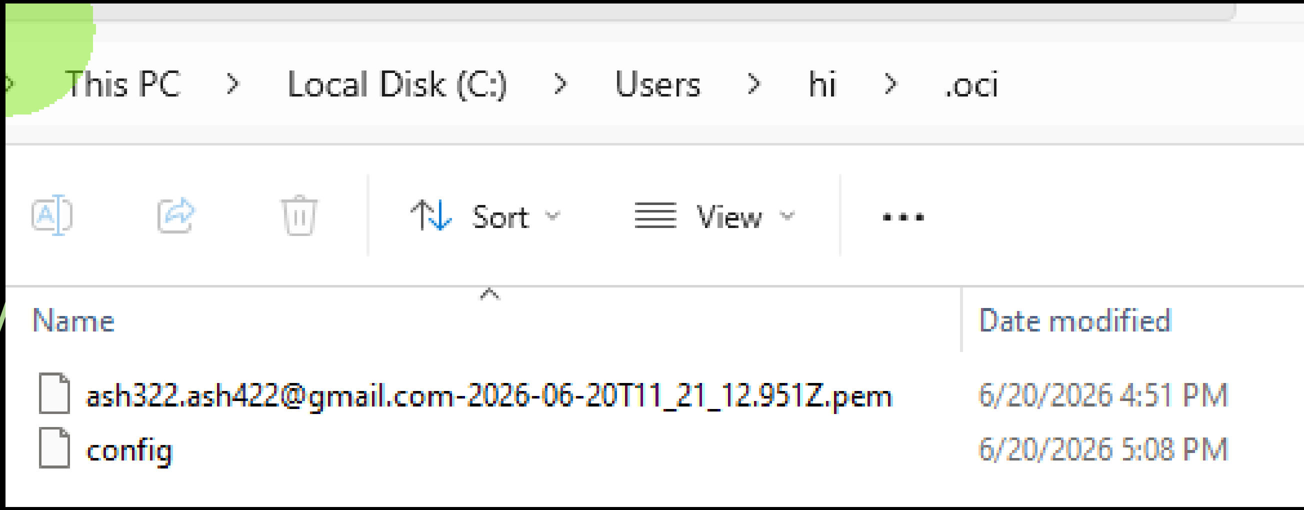
1

Configuration file preview [DEFAULT] user=ocid1.user.oc1..aaaaaaaaahu7lckuwznvwcebnv3gbodmbfl55hvcwyn7gxsmmaqzybq3kdwq fingerprint=c8:4c:f3:24:96:0a:94:e4... Copy

On Windows:

1. Create a directory named .oci inside your user profile folder

(e.g., C:\Users\YourName\.oci).



2. Move the your_key.pem in this **.oci** directory

3. Save your copied configuration block into a text file named **config** inside that folder.

4. In **config** file, change the **key_file** to path of your key file:
key_file=C:\Users\YourName\.oci\your_key.pem

```
[DEFAULT]
user=ocid1.user.oc1..aaaaaaaahu[redacted]f155hvcwyn7gxsmmaqzybq3kdwq
fingerprint=c8:4c:f3:24:96:0a:94:e4:cf:78:49:36:39:[redacted]
tenancy=ocid1.tenancy.oc1..aaaaaa[redacted]eutjopgcagugqrxyng4pn5m2ft6uoccpbd675uuq
region=ap-hyderabad-1
key_file=C:\Users\hi\.oci\ash322.ash422@gmail.com-2026-06-20T11_21_12.951Z.pem
```

```
!pip install oci
```

```
import importlib.metadata

# Fetch the version directly from the package metadata
oci_version = importlib.metadata.version("oci")
print(f"OCI version: {oci_version}")
```

```
OCI version: 2.179.0
```

Test the connection ¶

```
import oci

config = oci.config.from_file()

print("Connected Successfully")
print(config["region"])
```

```
Connected Successfully
ap-hyderabad-1
```

Let perform Sentiment Analysis

```python
import oci

config = oci.config.from_file()

ai_client = oci.ai_language.AIServiceLanguageClient(config)

text = "Oracle AI services are amazing."

document = oci.ai_language.models.TextDocument(
    key="1",
    text=text,
    language_code="en"
)

details = oci.ai_language.models.BatchDetectLanguageSentimentsDetails(
    documents=[document]
)

response = ai_client.batch_detect_language_sentiments(details)

print(response)
```

```
<oci.response.Response object at 0x000002F6B27D0710>
```

```python
print(response.data)
```


```
{
  "documents": [
    {
      "aspects": [
        {
          "length": 18,
          "offset": 0,
          "scores": {
            "Mixed": 0.00556711256520368,
            "Negative": 0.002052629611962227,
            "Neutral": 0.0,
            "Positive": 0.9923802578228341
          },
          "sentiment": "Positive",
          "text": "Oracle AI services"
        }
      ],
      "document_scores": {},
      "document_sentiment": "",
      "key": "1",
      "language_code": "en",
      "sentences": []
    }
  ],
  "errors": []
```


99.2% sure it is +

this is the Aspect

```python
text = """
Oracle AI services are excellent and easy to use.
However, the documentation is difficult to understand.
"""

document = oci.ai_language.models.TextDocument(
    key="1",
    text=text,
    language_code="en"
)

details = oci.ai_language.models.BatchDetectLanguageSentimentsDetails(
    documents=[document]
)

response = ai_client.batch_detect_language_sentiments(details)

print(response.data)
```

→ 2 aspects here

(cont...)

```
{
  "aspects": [
    {
      "length": 18,
      "offset": 1,
      "scores": {
        "Mixed": 0.11781786857949311,
        "Negative": 0.005587006879568341,
        "Neutral": 0.0,
        "Positive": 0.8765951245409384
      },
      "sentiment": "Positive",
      "text": "Oracle AI services"
    },
    {
      "length": 13,
      "offset": 64,
      "scores": {
        "Mixed": 0.007484954730957052,
        "Negative": 0.9916960680708166,
        "Neutral": 0.0,
        "Positive": 0.0008189771982264554
      },
      "sentiment": "Negative",
      "text": "documentation"
    }
  ],
```


1st aspect

2nd aspect

Note: *
Oracle gave
sentence/aspect
wise analysis